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Code and its output:

12:33 PM (<1s) 2: Load California Housing data

```
import pandas as pd
from sklearn.datasets import fetch_california_housing

# Load the California Housing dataset (target: median house value in $100k units)
housing = fetch_california_housing(as_frame=True)
pdf = housing.frame.copy()
TARGET = "MedHouseVal"
print(f"Shape: {pdf.shape}")
display(pdf.head(10))
```

```
Shape: (20640, 9)
```

Table

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Lor
1	8.3252	41	6.984126984126984	1.0238095238095237	322	2.55555555555555...	37.88	
2	8.3014	21	6.238137082601054	0.9718804920913884	2401	2.109841827768014	37.86	
3	7.2574	52	8.288135593220339	1.073446327683616	496	2.80225988700564...	37.85	
4	5.6431	52	5.81735159817351...	1.0730593607305936	558	2.547945205479452	37.85	
5	3.8462	52	6.281853281853282	1.0810810810810811	565	2.18146718146718...	37.85	
6	4.0368	52	4.761658031088083	1.1036269430051813	413	2.139896373056995	37.85	
7	3.6591	52	4.93190661478599...	0.9513618677042801	1094	2.12840466926070...	37.84	

12:33 PM (1s) 3: Exploratory data analysis

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
from sklearn.datasets import fetch_california_housing

# Reload data if lost after a %pip restart
if 'pdf' not in dir():
    housing = fetch_california_housing(as_frame=True)
    pdf = housing.frame.copy()
    TARGET = "MedHouseVal"

# Summary statistics and dtypes
pdf.info()
display(pdf.describe().T.reset_index().rename(columns={"index": "feature"}))

# Feature type table
feature_types = pd.DataFrame({
    "feature": pdf.columns,
    "type": ["numeric float (target)" if c == TARGET else "numeric float" for c in pdf.columns],
})
display(feature_types)

# Correlogram
fig, ax = plt.subplots(figsize=(9, 7))
sns.heatmap(pdf.corr(), annot=True, fmt=".2f", cmap="coolwarm", ax=ax)
```

12:33 PM (1s) 3: **Exploratory data analysis**

```
    fig, ax = plt.subplots(figsize=(9, 7))
    sns.heatmap(pdf.corr(), annot=True, fmt=".2f", cmap="coolwarm", ax=ax)
    ax.set_title("Feature correlation matrix")
    plt.show()

    # Relationship of key features with the target
    fig, axes = plt.subplots(1, 3, figsize=(18, 5))
    sample = pdf.sample(5000, random_state=42)
    sns.scatterplot(data=sample, x="MedInc", y=TARGET, alpha=0.3, ax=axes[0])
    axes[0].set_title("Median income vs house value")
    sns.scatterplot(data=sample, x="AveRooms", y=TARGET, alpha=0.3, ax=axes[1])
    axes[1].set_xlim(0, 15)
    axes[1].set_title("Average rooms vs house value")
    sc = axes[2].scatter(sample["Longitude"], sample["Latitude"], c=sample[TARGET], cmap="viridis", s=5)
    plt.colorbar(sc, ax=axes[2], label=TARGET)
    axes[2].set_title("House value by location")
    plt.tight_layout()
    plt.show()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 9 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   MedInc       20640 non-null  float64
 1   HouseAge     20640 non-null  float64
 2   AveRooms     20640 non-null  float64
 3   AveBedrms    20640 non-null  float64
 4   Population   20640 non-null  float64
 5   AveOccup     20640 non-null  float64
 6   Latitude     20640 non-null  float64
 7   Longitude    20640 non-null  float64
 8   MedHouseVal  20640 non-null  float64
dtypes: float64(9)
memory usage: 1.4 MB
```

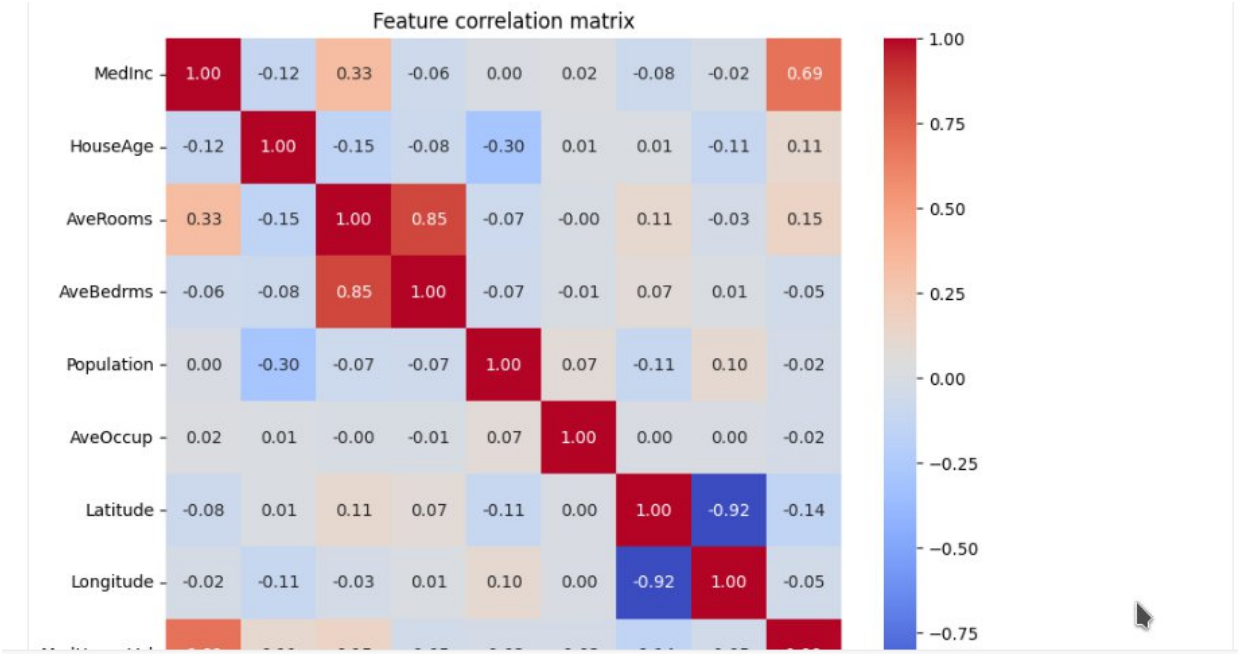
Table

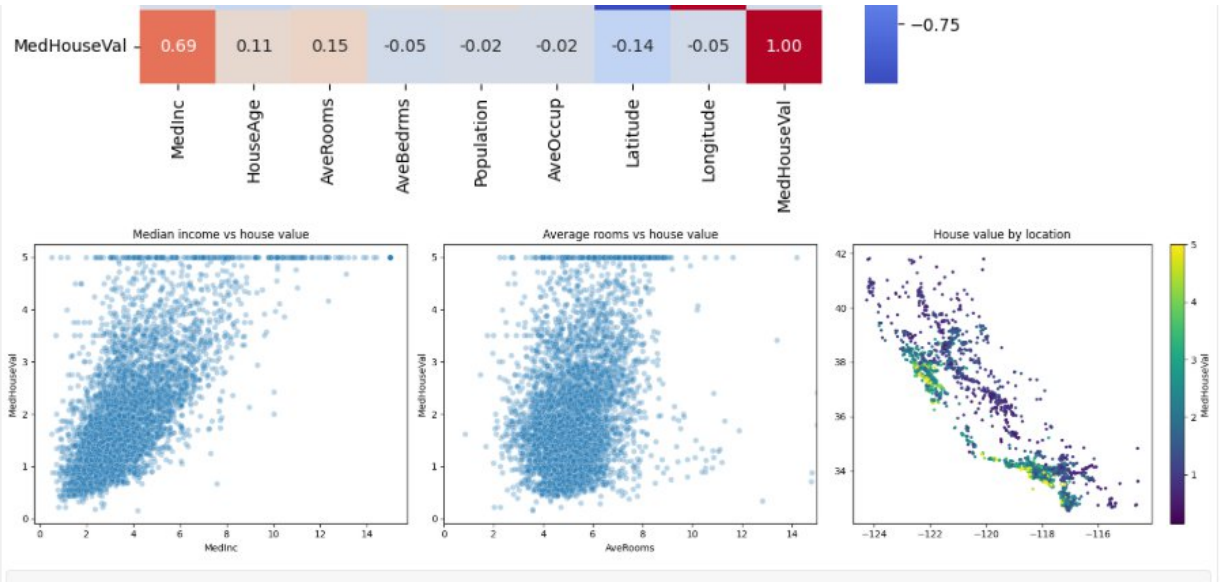
	feature	count	mean	std	min	25%	50%	75%
1	MedInc	20640	3.8706710029069766	1.89982171794526…	0.4999	2.56339999999999…	3.53479999999999…	
2	HouseAge	20640	28.639486434108527	12.58555761211165	1	18	29	
3	AveRooms	20640	5.428999742190376	2.47417313942431…	0.84615384615384…	4.440716235896959	5.229128787878788	6.0523
4	AveBedrms	20640	1.096675149606208	0.473910856795466	0.33333333333333…	1.006079046038478	1.048780487804878	1.0995
5	Population	20640	1425.4767441860465	1132.462121765341	3	787	1166	
6	AveOccup	20640	3.0706551594363742	10.3860495622136…	0.69230769230769…	2.42974114755357…	2.818115654360196	3.28226
7	Latitude	20640	35.63186143410853	2.13595239745711…	32.54	33.93	34.26	
8	Longitude	20640	-119.569704457364…	2.00353172350258…	-124.35	-121.8	-118.49	
9	MedHouseVal	20640	2.068558169089147	1.15395615874413…	0.14999	1.196	1.797	2.64724

Table

	feature	type
1	MedInc	numeric float
2	HouseAge	numeric float
3	AveRooms	numeric float
4	AveBedrms	numeric float
5	Population	numeric float
6	AveOccup	numeric float
7	Latitude	numeric float
8	Longitude	numeric float
9	MedHouseVal	numeric float (targ…

9 rows | 1.488s runtime





▶ ✓ 12:37 PM (16s) 4: **Install AutoML runtime package**

```
%pip install databricks-automl-runtime --quiet
%restart_python
```

```
Note: you may need to restart the kernel using %restart_python or dbutils.library.restartPython() to use updated packages.
```